

Linear systems and elimination

Uniqueness of the right inverse minimizing an induced p -to-2 norm

Difficulty: hard **Importance:** interesting to specialist

Status: Open

Rating rationale: Hard reflects a focused uniqueness question for a known norm minimizer; specialist impact is the characterization of generalized inverses under induced norms.

Problem statement

Let $1 \leq m < n$, let $A \in \mathbb{C}^{m \times n}$ have rank m , and let $2 < p < \infty$. For $X \in \mathbb{C}^{n \times m}$ define the induced norm

$$\|X\|_{p \rightarrow 2} = \sup_{y \in \mathbb{C}^m \setminus \{0\}} \frac{\|Xy\|_2}{(\sum_{i=1}^m |y_i|^p)^{1/p}}.$$

The Moore–Penrose inverse $A^\dagger = A^*(AA^*)^{-1}$ minimizes this norm among the right inverses X satisfying $AX = I_m$. Is it always the unique minimizer? Equivalently, must

$$AX = I_m, \quad X \neq A^\dagger \quad \implies \quad \|X\|_{p \rightarrow 2} > \|A^\dagger\|_{p \rightarrow 2}$$

hold for every such A, p, X ?

This is the direct-inverse uniqueness question in Dokmanić–Gribonval’s Remark 4.1. The endpoints and the separate objective $\|XA\|_{p \rightarrow 2}$ are outside this statement.

Connection to numerical linear algebra

The norm measures the worst Euclidean solution amplification for right-hand sides bounded in ℓ^p . Uniqueness determines whether another linear solver can match the Moore–Penrose inverse’s optimal amplification while changing its sparsity or structure.

References

1. I. Dokmanić and R. Gribonval, *Beyond Moore–Penrose Part I: Generalized Inverses that Minimize Matrix Norms*, arXiv:1706.08349v2 (2017), §2.1 and §2.4 (right inverses and $A^\dagger$); §4.4, Corollary 4.2(3) and Remark 4.1, manuscript p. 18 (PDF page 18). [Preprint and version history](#). [PDF](#).
2. I. Dokmanić and R. Gribonval, *Part II: The Sparse Pseudoinverse*, 2017, §2. Its entrywise-norm objective is different.

Earlier status check — 2026-09-08

Checked the complete v2 statement and version history on 2026-09-08; v2, dated 2017-07-13, remains the latest arXiv version of Part I. Searches combined the authors, the paper title, “induced norm”, “ p to 2”, “unique”, “uniqueness”, “generalized inverse”, and 2025/2026. The companion paper’s sparse-inverse uniqueness results concern entrywise norms and generic inputs, not this induced-norm claim for every full-row-rank matrix. No later proof or counterexample was located. This bounded search does not certify that no resolution exists elsewhere.

Audit update — 2026-09-10

The [author copy of Part I](#), Corollary 4.2(3) and Remark 4.1 on printed p. 18, explicitly separates minimality from the remaining uniqueness question for $2 < p < \infty$. Searches for later induced- p -to-2 uniqueness results found no resolution; Part II’s sparse-inverse objectives do not supply this missing assertion.